

A Xenon Hall-Effect Thruster for Smallsat Electric Propulsion: Scaling Model, Simulation Code, and Cost Analysis

Mayank Swaraj

Mayank Creations — Daily AeroSpace Concept Series, Day 4

Patna, Bihar, India · July 2026

[linkedin.com/in/mayank-swaraj-4b84b6410](https://www.linkedin.com/in/mayank-swaraj-4b84b6410) · github.com/mayankswaraj18cr-cmd

Abstract — Hall-effect thrusters accelerate ionized xenon propellant across a radial magnetic field crossed with an axial electric field, trading the very high thrust of chemical propulsion for specific impulse an order of magnitude higher — enabling orbit-raising and stationkeeping missions that would need far more propellant mass with any chemical system. This paper presents an energy-limited ion-acceleration scaling model, the accompanying browser simulator that turns thruster design (voltage, mass flow) and spacecraft constraints (power budget, propellant load, ΔV target) into a mission-feasibility check, and a cost analysis in Indian Rupees spanning an engineering-model thruster through flight qualification. ISRO has already flown xenon-based electric propulsion operationally on GSAT-9 and GSAT-4, and Hall thrusters are the propulsion backbone of Starlink-class LEO constellations, making this one of the most commercially relevant concepts in the series.

1. Introduction

Every propulsion decision trades thrust for efficiency. A Hall-effect thruster (HET) sits at the efficiency extreme of that trade: specific impulse of 1,500–2,500 s (5–10× a chemical bipropellant engine) at thrust levels of only tens to hundreds of millinewtons — far too little to launch anything, but more than enough to raise or maintain an orbit over weeks to months using a fraction of the propellant mass chemical propulsion would need. This is why HETs have become the default choice for LEO constellation stationkeeping and orbit raising (Starlink), and why ISRO has flown xenon ion propulsion operationally on GSAT-9 and GSAT-4 for north-south stationkeeping. This paper documents the underlying scaling physics, a browser simulator that lets a team check whether a given thruster design fits their spacecraft's power budget and propellant allowance, and a transparent INR cost model.

2. Thruster Scaling Model

2.1 Ion acceleration and exhaust velocity

Each singly-ionized xenon ion is accelerated through the discharge voltage V , gaining kinetic energy equal to its charge times the voltage drop:

$$\frac{1}{2} \cdot m_{ion} \cdot V_e^2 = q \cdot V \quad \blacksquare \quad V_e = \sqrt{(2qV/m_{ion})}$$

with elementary charge $q = 1.602 \times 10^{-19}$ C and xenon ion mass $m_{ion} = 2.18 \times 10^{-25}$ kg. At $V = 300$ V this gives an ideal exhaust velocity of roughly 21 km/s — for comparison, the best chemical rocket engines in this series (Day 1's hybrid motor) exhaust around 2–2.3 km/s, an order-of-magnitude difference that is the entire reason electric propulsion exists.

2.2 Thrust, beam current, and required power

Thrust is the momentum flux of the ion beam, discounted by an overall thruster efficiency η that accounts for incomplete ionization, beam divergence, and electrical losses ($\eta \approx 0.55$ is representative of flight-proven mid-power HETs):

$$F = \eta \cdot \dot{m} \cdot V_e$$

Beam current follows from the particle flow rate of singly-charged ions:

$$I_{beam} = \dot{m} \cdot q / m_{ion}$$

and total electrical power required is the beam power ($I \cdot V$) divided back by the same efficiency factor:

$$P = I_{beam} \cdot V / \eta \quad I_{sp} = F / (\dot{m} \cdot g_0)$$

These three equations are the entire design space of a Hall thruster at the system level: raise voltage V and I_{sp} rises (better propellant economy) but so does the power needed for a given thrust; raise mass flow $\dot{m}$ and thrust rises but I_{sp} falls. The simulator's power-budget check — comparing required P against the spacecraft's actual available power — is usually the binding constraint for a smallsat, more often than propellant mass.

2.3 Mission ΔV integration

The simulator integrates the rocket equation numerically rather than using the closed-form Tsiolkovsky relation, because when the required electrical power exceeds the available power budget the thruster can only run at a duty cycle (available power / required power), reducing effective average thrust and mass flow. At each 6-hour timestep the model computes instantaneous acceleration $a = F_{eff}/m$, accumulates ΔV , and depletes both spacecraft wet mass and propellant reserve until either the ΔV target is reached or the propellant load is exhausted — whichever comes first.

2.4 Model limitations

- $\eta = 0.55$ is a fixed representative efficiency; real HETs vary 45–65% efficiency depending on power class and design maturity, and efficiency itself varies with operating point (voltage, mass flow) rather than being constant.
- No magnetic-field topology, discharge-channel erosion, or plume divergence angle is modelled — this is a system-level power/thrust/ I_{sp} scaling tool, not a plasma-physics simulation.
- Thruster lifetime (limited by discharge-channel wall erosion, typically thousands of operating hours for flight HETs) is not modelled; total propellant throughput in a real mission is often lifetime-limited before it is propellant-mass-limited.
- Orbital mechanics beyond simple ΔV accumulation (e.g. actual LEO drag profile, eclipse power availability, plane-change ΔV penalties) are outside this simplified model's scope.

3. Interactive Simulator (accompanying HTML file)

The companion file **day4_hall_thruster.html** is a single self-contained, mobile-only page implementing the model above with six sliders (voltage, mass flow, power budget, spacecraft mass, propellant load, ΔV target). It gives an immediate **Y/N** verdict on whether the chosen thruster design fits the stated power budget, and charts ΔV accumulation against a dashed target line — turning an otherwise abstract sizing exercise into something a team can iterate on interactively before committing to hardware.

3.1 Core scaling and mission-loop code (excerpt)

```
const Ve = Math.sqrt(2*Q*M_ION); // ideal exhaust velocity
const thrust = ETA * mdot * Ve; // N
const beamCurrent = mdot * Q / M_ION; // A
const powerNeed = (beamCurrent * V) / ETA; // W

const duty = Math.min(100, powerBudget/powerNeed*100);
const effMdot = mdot * (duty/100);
const effThrust = thrust * (duty/100);

for (let i = 0; i < 2000; i++) { // 6-hour steps
  const a = effThrust / m;
```

```

dvSoFar += a * dtstep;
propUsed += effMdot * dtstep; m -= effMdot*dtstep; t += dtstep;
if (propUsed >= propLoad) { exhausted = true; break; }
if (dvSoFar >= dvTarget) { reached = true; break; }
}

```

4. Cost Analysis — INR

Xenon spot pricing in 2026 ranges roughly US\$1,100–2,930/kg depending on purity grade and region (Russia at the low end, US/EU spot at the high end); this paper uses ■1,50,000/kg (≈US\$1,700/kg at ■87/US\$) as a representative mid-range figure for aerospace-grade xenon. ISRO's own procurement of high-purity xenon for GSAT electric propulsion testing (beginning 2018) confirms this is a live, real procurement category for the Indian space program, not a hypothetical one.

4.1 Engineering-model thruster, breadboard/lab scale

Item	Basis	Cost (■)
Xenon propellant (2–10 kg typical lab load)	■1,50,000/kg	3,00,000 – 15,00,000
Power processing unit (PPU), lab-grade	scaled to 100–1000 W	1,50,000 – 8,00,000
Thruster head (discharge channel, magnetic circuit)	in-house or COTS EM unit	3,50,000 – 9,00,000
Hollow cathode + neutralizer	1 unit	1,20,000 – 3,00,000
High-pressure xenon tank + regulator	1 set	90,000 – 2,50,000
Vacuum chamber test time (thrust stand access)	per test campaign	4,00,000 – 12,00,000

Table 1 — Engineering-model (EM) thruster suitable for lab characterization and vacuum-chamber thrust-stand testing, not flight.

EM-scale total: ≈ ■14,10,000 – ■49,50,000, roughly ■0.14 – 0.5 crore — already crore-adjacent even at lab scale, because xenon and vacuum-chamber test time are both intrinsically expensive relative to the chemical-propulsion prototypes in Days 1–2 of this series.

4.2 Flight-qualified unit

Program element	Basis	Cost (■)
Flight-grade thruster head + magnetic circuit qualification	vibration, thermal-vac, life test	40,00,000 – 90,00,000
Flight PPU (radiation-tolerant, redundant)	space-grade electronics	35,00,000 – 80,00,000
Xenon flow control + tank (flight-qualified, higher pressure rating)	COTS flight hardware	15,00,000 – 35,00,000
Life-test campaign (1,000+ hour endurance run)	thousands of vacuum-chamber hours	25,00,000 – 60,00,000
Integration, harnessing, documentation, contingency	program-wide	15,00,000 – 30,00,000

Table 2 — Indicative flight-qualified Hall-thruster unit cost, consistent with published international small-satellite electric-propulsion module pricing (~US\$300,000–900,000 per qualified unit).

Flight-qualified unit total: ≈ ■1,30,00,000 – ■2,95,00,000, roughly ■1.3 – 3 crore per qualified thruster unit — genuinely crore-scale even for a single unit, driven by life-testing (thruster erosion life is the dominant qualification risk for HETs) rather than by materials cost.

5. Space History — Random Date

December 21, 1971 — the Soviet Meteor-series satellite carried the first operational Hall-effect thruster (SPT-50, developed at Fakel Design Bureau) to orbit, beginning a propulsion lineage that today includes ISRO's GSAT electric-propulsion satellites and the Hall thrusters used across the Starlink constellation for

6. Conclusion

Hall-effect propulsion is the highest-Isp, lowest-thrust concept in this series so far, and the first where cost is dominated by the propellant and test infrastructure (xenon spot price, vacuum-chamber life testing) rather than by fabrication. Even an engineering-model unit reaches ■0.14–0.5 crore before any flight qualification, and a qualified flight unit runs ■1.3–3 crore — a useful contrast against Day 1's chemical hybrid motor, where the prototype stayed in the lakhs and only the full flight program reached crore scale. The underlying lesson across the series so far: what pushes a concept into crore-scale spend is qualification testing and, for electric propulsion specifically, the propellant and vacuum-facility costs themselves.

References

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